

BASIT ALI

Principal Big Data Engineer

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EXECUTIVE SUMMARY

Principal Big Data Engineer with 9+ years architecting and leading enterprise-scale data platforms across healthcare, telecom, government, retail, and IoT. Owns end-to-end technical direction for large data platforms, including architecture, governance, team standards, and cost efficiency, spanning on-premises Cloudera/Hadoop, Apache Iceberg lakehouses, Microsoft Fabric, and multi-cloud (AWS/Azure/GCP) environments. Sets engineering standards and mentors engineers, while partnering with stakeholders to translate business needs into production-grade, real-time data systems. Track record of double-digit cost and performance gains at national/enterprise scale.

CORE ARCHITECTURE COMPETENCIES

- Data Lake, Lakehouse (Apache Iceberg, Hudi), Data Mesh, Data Fabric & Medallion Architecture
- Cloud & On-Prem Data Architecture, Modeling, Governance & Metadata Management
- ETL/ELT Architecture, Pipeline Automation & Real-Time Streaming at Scale
- Data Integration, Federation, Virtualization & Master Data Management
- Technical Leadership: standards-setting, mentorship, and cross-team architecture ownership
- AI/ML & Generative AI-Enabled Data Platform Architecture

TECHNICAL SKILLS

- Big Data & Analytics: Cloudera Data Platform (CDP/CDH) | Apache Spark | Apache Iceberg | Apache Hudi | Kafka | Airflow | Databricks | Hadoop | HDFS | MapReduce | Hive | Impala | Neo4j | Power Apps
- Real-Time Streaming: Apache Kafka | Kafka Streams | Spark Streaming | Apache Flink | Apache NiFi | AWS Kinesis Firehose | Event-Driven Architecture
- Cloud Platforms: Microsoft Fabric | Azure (ADF, Synapse Analytics, Databricks, Functions, DevOps) | AWS (EMR, Glue, Lambda, Kinesis, Redshift, S3, Step Functions) | GCP
- Programming: Python | PySpark | Java | SQL | Node.js | Angular
- Databases: HP Vertica | Teradata | Redshift | Snowflake | BigQuery | Azure Synapse | MongoDB | Elasticsearch | Redis | PostgreSQL
- DevOps & Tools: GitHub | Azure DevOps | Jenkins | TFS | Trello | Jira | MS Visio
- Artificial Intelligence: Generative AI | RAG Applications | Agentic AI | Deep Learning | Machine Learning

WORK EXPERIENCE

Senior Big Data Engineer, Technical Lead

February 2025 – Present

Lean Business Services, Riyadh, Saudi Arabia

- Own end-to-end architecture and technical direction for a national-scale Patient Profile Platform on Apache Spark and Iceberg-managed lakehouse tables, consolidating clinical data from hospitals across the Kingdom into a single enterprise source of truth for millions of patient records.
- Designed patient linkage/identity-resolution algorithms to match and merge records across disparate hospital systems, eliminating duplication and enabling a 360-degree longitudinal patient view that directly informs clinical and operational decision-making.
- Set the ingestion framework standard for the platform built a metadata-driven onboarding tool (Streamlit + Python) enabling SQL, MySQL, Oracle, and NoSQL sources to be configured without manual coding, cutting new-facility onboarding effort significantly.
- Architected real-time pipelines (Spark Streaming, Kafka, Apache NiFi) achieving sub-5-second data availability for critical clinical and operational decisions across all connected facilities.

- Optimized long-running Oracle extraction and Iceberg merge/write workflows (partition pruning, merge-on-read strategy), cutting batch processing windows by 40% and improving daily-load reliability at multi-terabyte scale.
- Defined and enforced data governance, quality-control, and privacy standards across all pipelines, ensuring compliance with national healthcare data protection regulations.
- Drop 30% infrastructure cost reduction by redesigning event-driven architectures and decommissioning redundant/legacy pipelines, while maintaining platform SLAs.

Senior Big Data Engineer

September 2023 – January 2025

Ai-Elements, Riyadh, Saudi Arabia

- Led architecture and delivery of multi-source pipelines (Spark, NiFi, Airflow) ingesting from 10+ enterprise systems into a centralized analytics platform, enabling near real-time operational insight for business stakeholders.
- Partnered directly with data science and product leadership to translate business requirements into Airflow-orchestrated workflows, cutting manual processing effort by 25% and accelerating time-to-insight.
- Directed end-to-end pipeline delivery across Airflow, NiFi, and Spark, and stood up terabyte-scale Cloudera/Hadoop batch workflows achieving 99.9% reliability with full data-security compliance.
- Set engineering standards for the team, including pipeline monitoring, code review, and data-quality checks, cutting production incidents by 20% and reducing technical debt by 30% through Clean Code refactoring.
- Built a Cloudera-based generative AI/ML proof-of-concept using Spark and Hadoop to validate large-scale model training and inference performance for future platform investment.

Senior Big Data Engineer

July 2019 – September 2023

Systems Limited, Islamabad, Pakistan

- Architected Microsoft Fabric Medallion Architecture and enterprise-scale pipelines (Spark, Hadoop), improving ingestion, transformation, and analytics performance across telecom, retail, and government clients.
- Built distributed ETL frameworks (Spark, Python, AWS EMR) processing 10+ TB of data daily, improving data accuracy by 35% and enabling actionable insight across multiple client domains.
- Engineered multi-service AWS data platforms (EMR, Glue, Lambda, Kinesis, Step Functions, CloudFormation, RDS, S3, Redshift) supporting both batch and real-time workloads on scale.
- Delivered a real-time streaming platform (Kafka, Spark Streaming, Apache Flink) with sub-second latency for fraud detection and live operational monitoring across telecom and retail domains.
- Built multi-cloud pipelines spanning Azure and AWS (ADF, Azure Functions, Queues, Redis, Databricks) for near real-time ETL and cross-platform data movement.
- Applied Clean Code, SOLID, and TDD practice refactor legacy systems, cutting technical debt by 30% while mentoring junior engineers on code quality standards.
- Implemented Jenkins-based CI/CD to orchestrate Spark jobs and infrastructure deployments, and automated SQL-based data validation to support QA verification cycles.

Big Data Developer

February 2018 – June 2019

Metis International Private Limited, Islamabad, Pakistan (incl. Dubai Future Accelerators, onsite)

- Built a web-based Big Data portal (Python, Django) and Neo4j graph visualizations to support enterprise data management and analysis.
- Delivered social media analytics (Python) for crime, hate-speech, and sentiment analysis, and geolocation analysis supporting a national Anti-Narcotics Department initiative.
- Developed Angular 2/5 front-end features include real-time alerts, live vehicle tracking, geofencing, and fuel/behavior analytics dashboards.

LEADERSHIP & MENTORSHIP

- Set technical direction and engineering standards across multiple data platform teams, driving adoption of code review, CI/CD, and data-quality practices.
- Mentor engineers on distributed systems design, Spark/Iceberg performance tuning, and Clean Code practices.
- Recognized as Team of the Year (2021) and Star Performer (2020, 2021) at Systems Limited for technical leadership and delivery impact.

CERTIFICATIONS & TRAINING

- Microsoft Certified: Azure Data Engineer Associate
- Cloudera Technical Expert (CTE) Accreditation
- Cloudera Technical Specialist (CTS) Accreditation
- Cloudera Technical Professional (CTP) Accreditation
- Future Leadership Program, Lahore University of Management Sciences (LUMS)

EDUCATION

Bachelor of Science in Software Engineering

International Islamic University, Islamabad, Pakistan, February 2018